Computers & Portable Tech

Late 1950s and beyond

Computer Music

- Bell Labs & Max Mathews 1957 (MUSIC I)
- IBM computers in the 1950s
- IRCAM – computer music research, started by Boulez but still in use today, RAVE model studies currently
- Today's Max/MSP named after Max Mathews

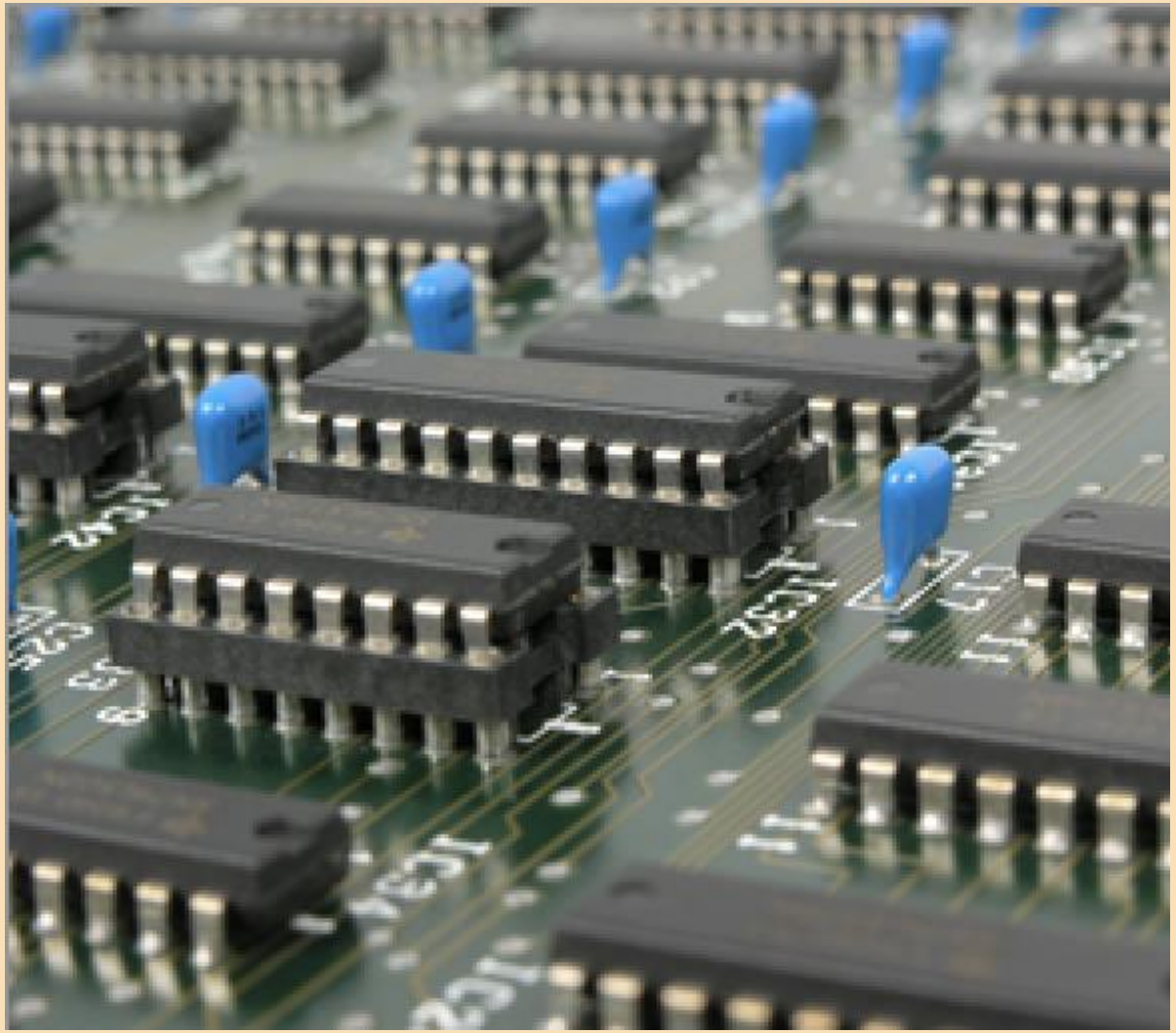
Computer Music Composers & Developers

Compositions using computer programming, often using synthesis concepts or computer-aided stochastic (probability) systems.

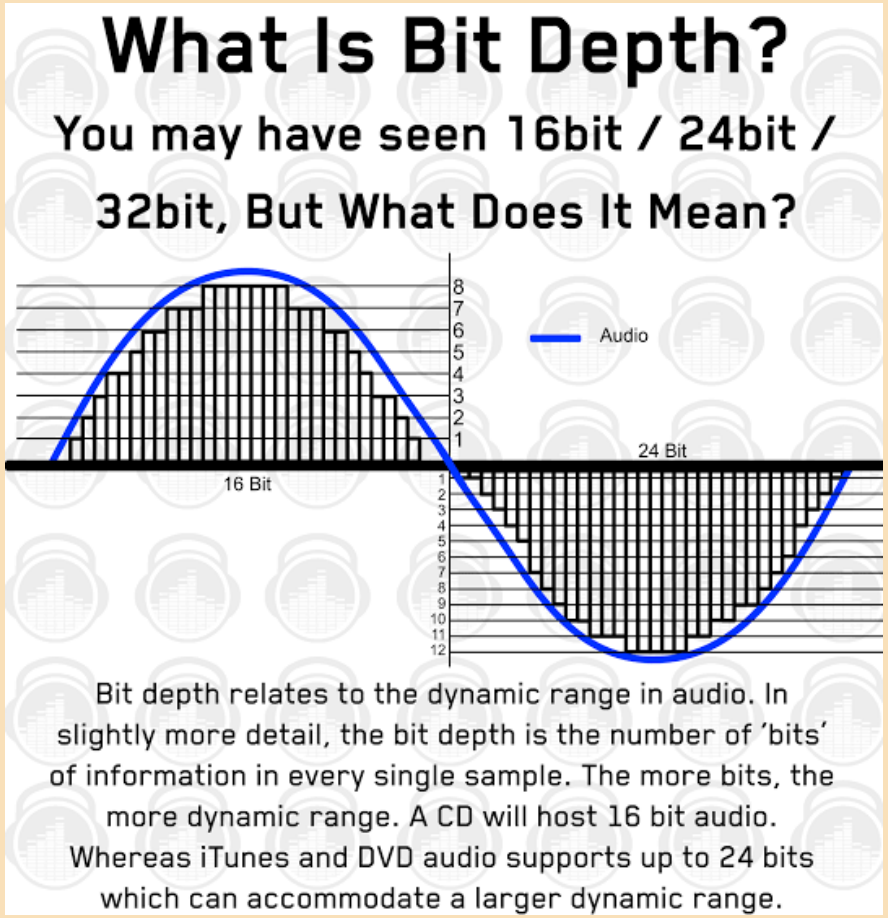
- Lejeren Hiller - Iliac Suite string quartet (1957)
- Iannis Xenakis - Metastaseis (1953, stochastic) & UPIC (1977)
- John Chowning (FM synthesis, spatialized sound)
- Laurie Speigel (Used GROOVE by Mathews)
- Paul Lansky - Idle Chatter (Princeton)
- Barry Vercoe (CSound, still in use today)
- Miller Puckett (IRCAM & invented Max/MSP)

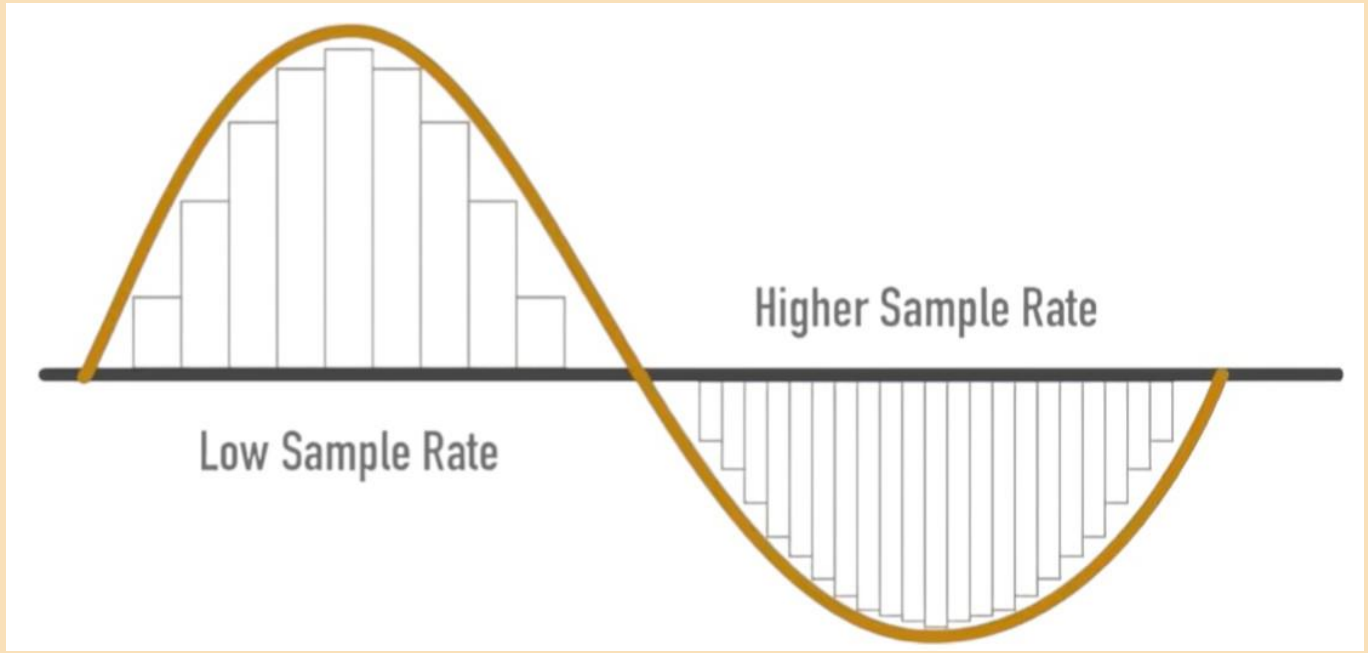
Transistors began to replace tubes mid-1960s with a slow development through the 1950s

Late 1960s, integrated circuits and microprocessors began to replace transistors



Digital Signal Processing (DSP) vs. analog sound of synths, tape, and records





DX7 FM synthesis very popular in the '80s with pop musicians

FM sample presets but can also do the FM yourself

Made Stanford a lot of money, John Chowning patent

Discovered FM in 1967, patent with Yamaha 1973, DX7 1983

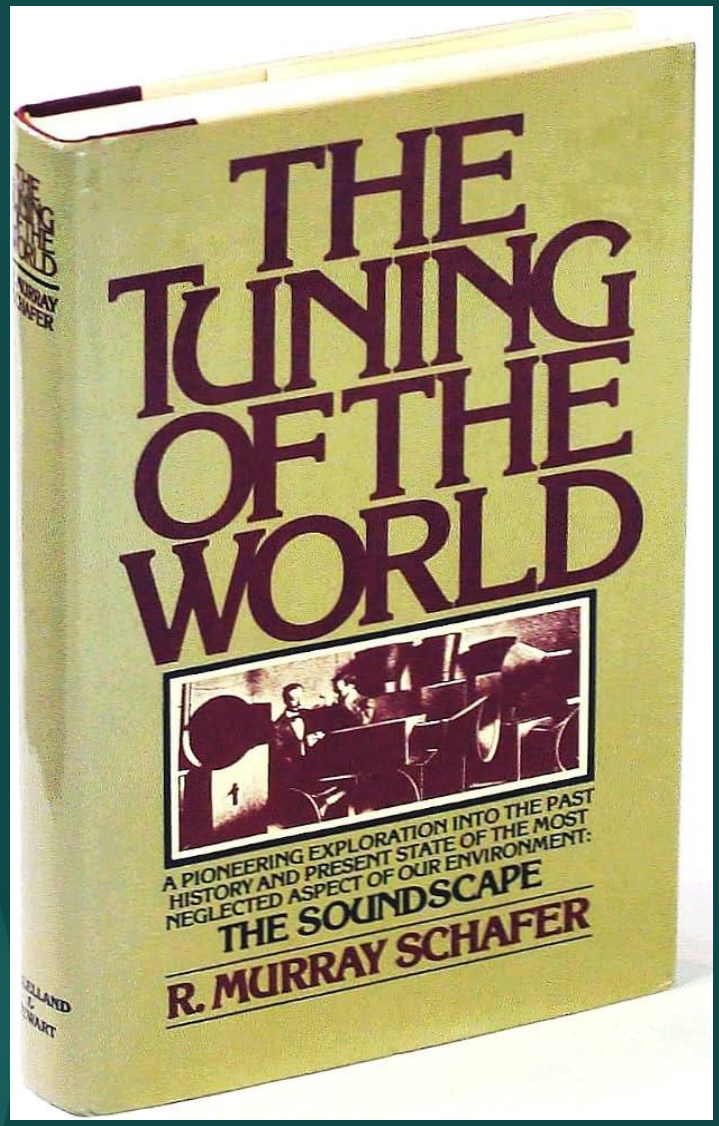


Field Recording & Portable Tech.

- R. Murray Schaefer
- Hildegard Westerkamp
- Luc Ferrari
- David Dunn - https://news.ucsc.edu/2017/02/bark-beetles-dunn/

- Acoustic ecology, 1960s-1970s beginnings
- Environmental sounds
- Urban landscapes
- Acoustics & architecture




THE TUNING OF THE WORLD
A PIONEERING EXPLORATION INTO THE PAST HISTORY AND PRESENT STATE OF THE MOST NEGLECTED ASPECT OF OUR ENVIRONMENT:
THE SOUNDSCAPE
R. MURRAY SCHAFER

Sound Art & Installation

Baschet Brothers: Structures Sonores

https://youtu.be/eco2rzDvxoM?si=jhi557Ap6eP9wnbf



Harry Bertoia

- https://harrybertoia.org/sonambient-sound-sculptures/
- https://youtu.be/a9EF_zO5ihs?si=ZmzR-5JIKkvL10Qc



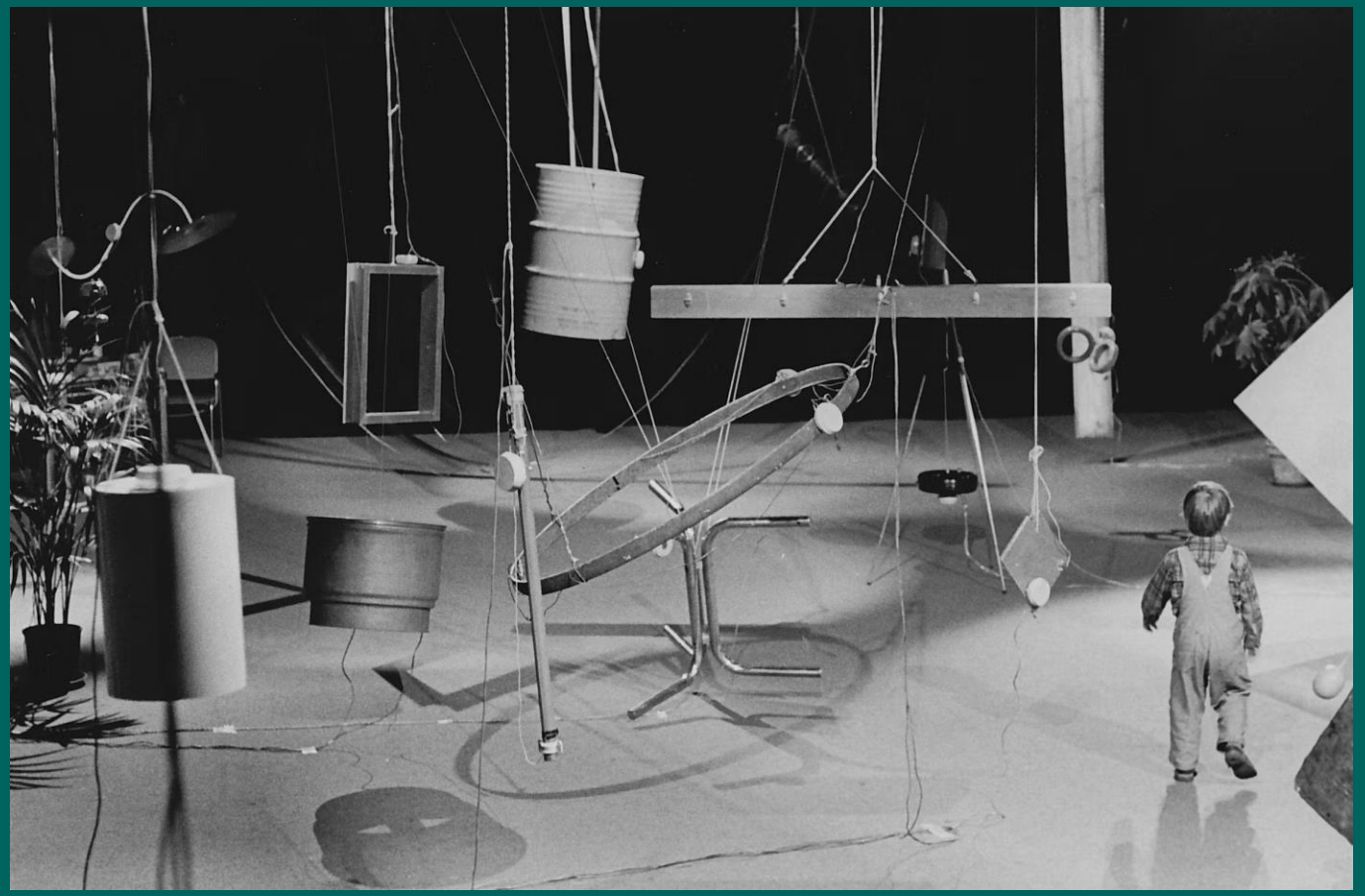
Jean Tinguely

- https://youtu.be/uESfCiePA9o?si=TpLeOFWYNmB2syW4



David Tudor: Rainforest

- https://www.moma.org/magazine/articles/166
- https://open.spotify.com/track/2K1eaNXIrlVXFtPQDmo9gF?si=dfb26504ff8c4fcc



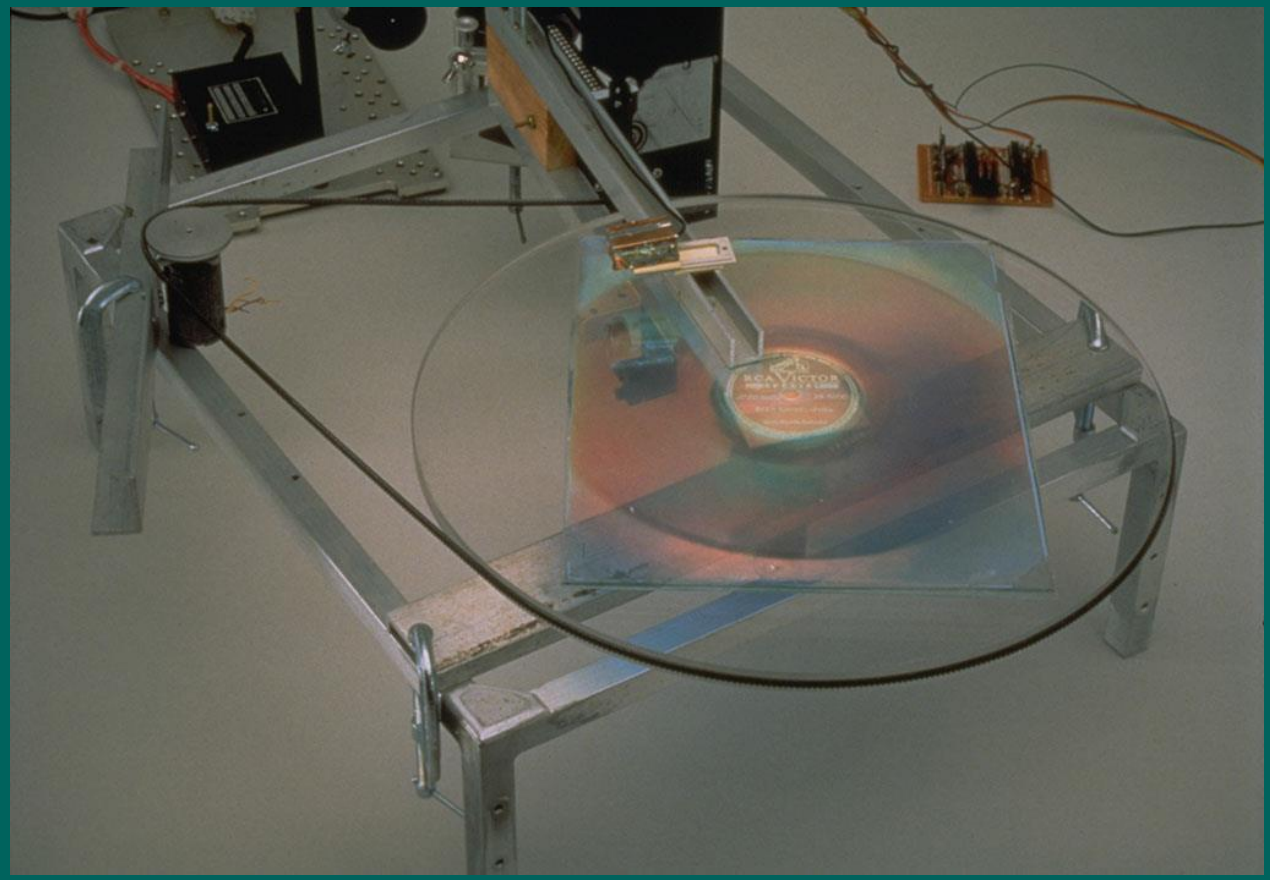
Bill Fontana

- https://www.resoundings.info/new-page-4
- Often recorded one environment and played it in a different one
- Environmental sound art becomes more prevalent in the 70s-90s due to technology and environmental issues



Paul DeMarinis

- https://pauldemarinis.org/EdisonEffect.html



Janet Cardiff

- 40 Part Motet
- https://youtu.be/38ORiaia9r8?si=01JiOZ9LhPD7MxBm



Tristan Perich

- Microtonal Wall
- 1-bit symphony
- https://vimeo.com/4522541 2
- https://youtu.be/fodArA6dI cs?si=xg1bdbSha4bl7QmE



Zimoun

- https://youtu.be/YqMyOPCQwS0?si=DTxfUSxZqHf4KEnL
- https://youtu.be/VAzObWw4ezo?si=iFAuiED4X8OTW9z0



Live Electronics & Performance

1960s-now

AMM (Cornelius Cardew) - 1965

David Tudor (friends with Cage, 4'33" premiere)

Robert Ashley

Alvin Lucier

League of Automatic Music Composers (late '70s)

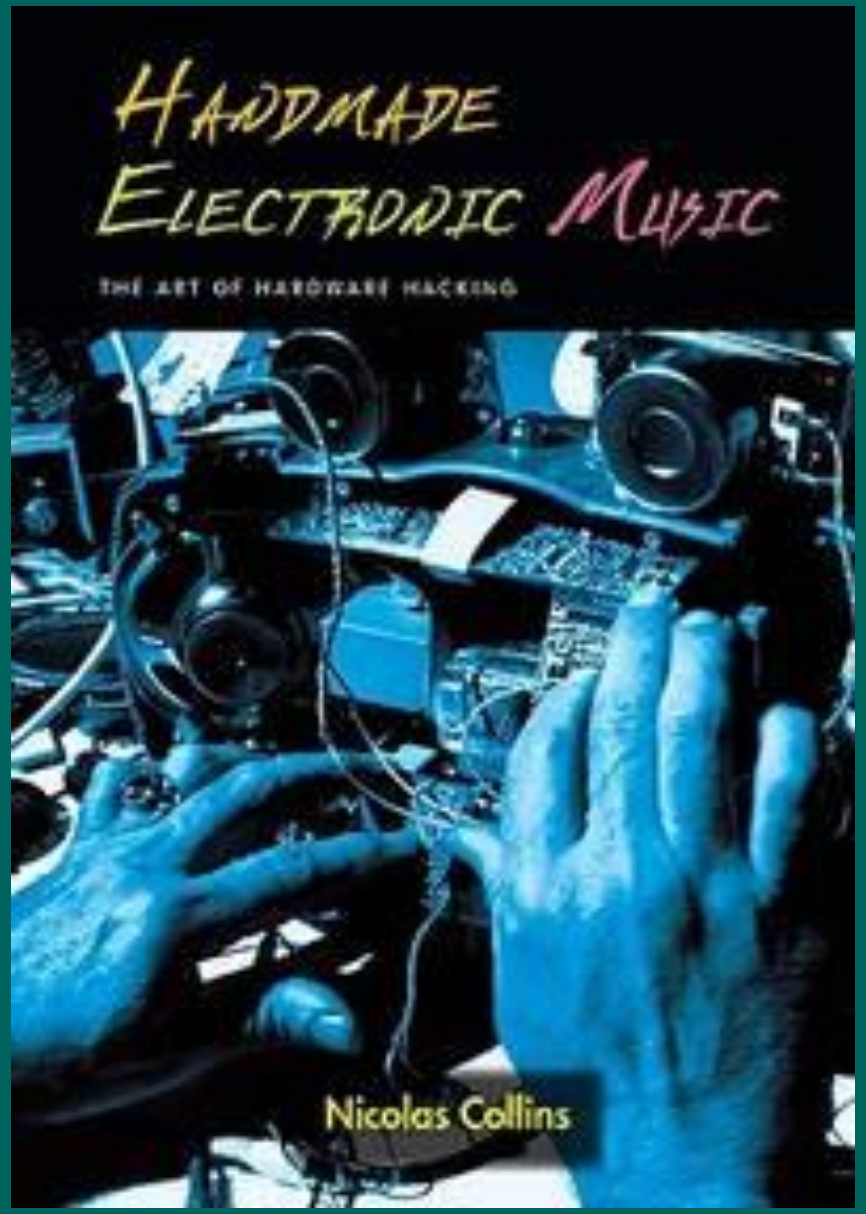
The Hub (computer music ensemble, 1986)

Maria Chavez (many other turntablists, as well)

And so many more!

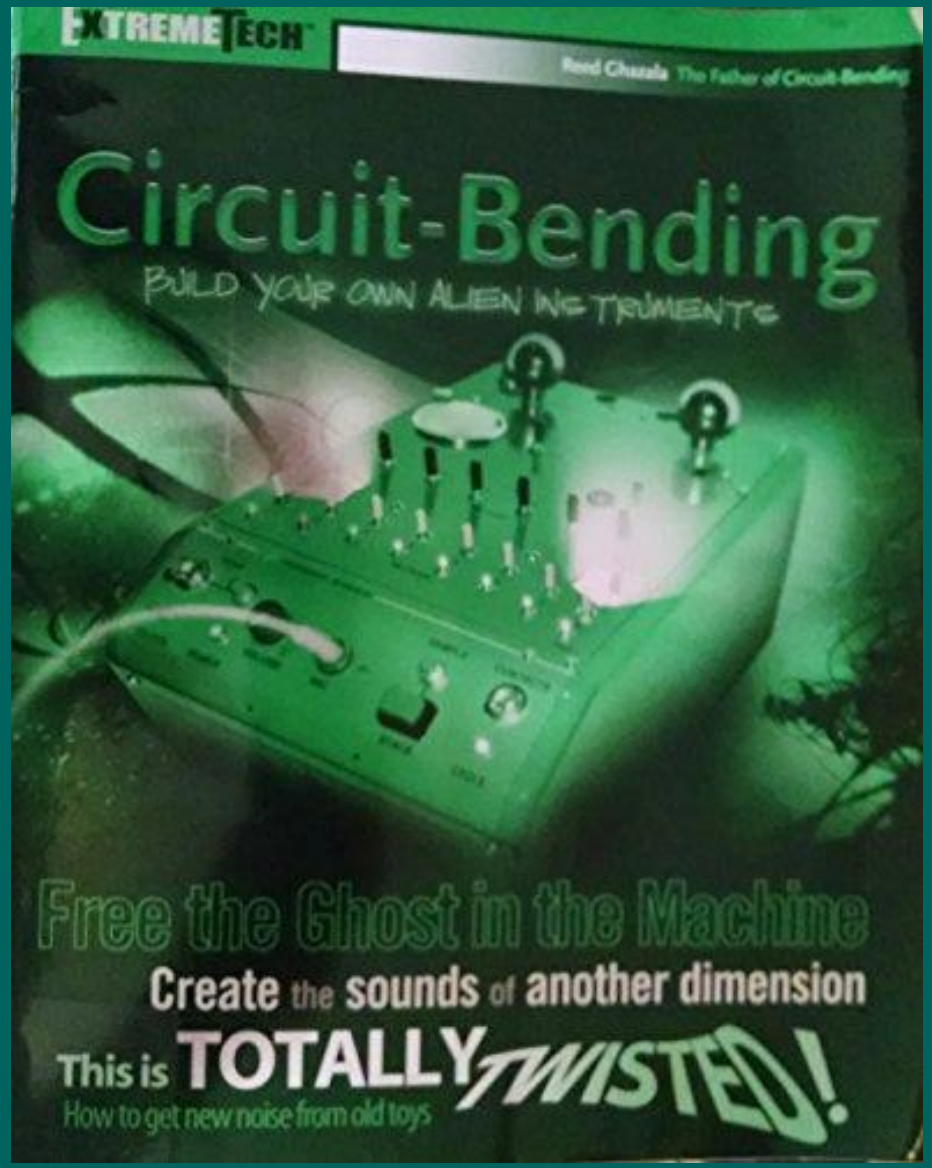
Nic Collins

- Hardware hacking audio and consumer electronics
- DIY electronics
- https://youtu.be/LF0yr0DKnGE?si=wC3gakSiAR3EaxQf
- https://youtu.be/I-RoQeW9ano?si=-R-Y8Y5t4j7UExd_
- https://youtu.be/AYFm9KifL5E?si=BTvRjR-QYmloOBsl



Reed Ghazala

- Circuit bending & hardware hacking
- https://youtu.be/KHDL9iGxDPM?si=L7CCahNPhwLDc7tb



Rie Nakajima

- Contemporary performer & sound artist
- https://youtu.be/1f6vbThPO2w?si=PO0kkDtQLoJ3JK0O
- https://youtu.be/dbB9Pn6nduM?si=yN5OvXaogvmVG7_d



Ryoko Akama

- Performer & sound artist
- https://vimeo.com/247511818
- https://ryokoakama.com/installation-page

